

William L Zhang

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Education

Harvard University

S.B. Candidate in Mechanical Engineering, Secondary in Computer Science

Cambridge, MA

May 2028

- Relevant Coursework: Mechanical Systems, Computer-Aided Machine Design, Mechanics of Solids, Circuits/Electronics, Data Structures & Functional Programming, Linear Algebra

Fudan University

Summer Study Abroad Program on the History of U.S. China Relations

Shanghai, China

June 2025 – August 2025

Technical Skills

Software & Robotics: Python, C, OCaml, MATLAB, PyTorch, dora-rs, MuJoCo, Inverse Kinematics, Motion Retargeting, CAN Bus, URDF

Mechanical & Fabrication: SolidWorks, FEA, GD&T, DFM/DFA, Tolerance Stack-Up, 3D Printing, CNC Milling, Manual Milling, Lathe, Drill Press

Engineering Experience

AprilRobotics

Robotics Engineering Intern

Zürich, Switzerland

May 2026 – August 2026

- Engineered a custom 3D-printed mechanical adapter and URDF model to mount an ORCA hand onto an OpenArm, integrating inverse kinematics for whole-arm control and virtual testing in MuJoCo before live retargeting
- Trained transformer-based manipulation policies on teleoperated data, encoding multi-camera visual and proprioceptive inputs to predict robot action sequences, achieving 90% success across two dexterous manipulation tasks
- Optimized a hand motion-retargeting pipeline mapping human hand poses to a 17-DoF robot hand, redesigning the joint-mapping algorithm to improve tracking efficiency and joint range of motion by over 80%
- Designed and assembled custom ESP32-based PCBs for robotic wrist cameras, applying reflow-oven soldering and surface-mount (SMD) rework to integrate MIPI-CSI camera sensors over USB

Harvard College Engineers Without Borders

El Mirador Team Member

Boston, MA

October 2025 – Present

- Develop EPANET visualizations for a solar-pumped water distribution system serving an underserved community in El Mirador, Guatemala, supporting an early-stage plan for household water access
- Contribute to decision-matrix evaluation of proposals, comparing design options and components such as pumps, tanks, pipelines, and household connections with a preliminary \$77.6K construction budget

ES-51 Project: Computer-Aided Machine Design – Turf Wars Robot

Fall 2025

- Designed and prototyped a competition robot that placed different objects on tiered shelves in a 3-minute match, producing a full CAD assembly and GD&T manufacturing drawings to guide fabrication; team placed 2nd of 8
- Fabricated custom components, including intake gears, through CNC milling, turning, broaching, laser cutting, and 3D printing, iterating through multiple prototypes to reduce object pick-up time from 23 seconds to 12 seconds

Leadership Experience

Harvard Undergraduate Association for U.S. Cross-Cultural Relations

Director of Business Pioneering Challenge (BPC), Previous Junior Director of HSYLC

Boston, MA

April 2025 – December 2025

- Organized HSYLC (Harvard Summit for Young Leaders in China), Asia's largest student-run conference, promoting liberal arts education to 1,000+ top-performing high school students across four provinces
- Designed and taught a week-long seminar on nanotechnology and quantum computing, introducing college-level STEM concepts to high school students through daily lectures and interactive discussions and activities
- Spearheaded conference programming and student life, leading academic panels, office hours, house events (Olympics, Trashion Show), extracurricular activities, and coordinating staff to deliver a nine-day transformative experience

Asian American Brotherhood

President

Boston, MA

March 2025 – Present

- Lead a 39-person organization, overseeing recruitment and year-round programming including homeless shelter and Chinatown outreach, cultural initiatives, and brotherhood outings
- Run weekly check-ins and facilitate discussions including State of the Brotherhood and deliberations, reinforcing positive social dynamics across the organization and resolving inter-member conflicts